9. Consider a circle with centre O and a $1\,\text{cm}$-long radius. Let P be a point outside this circle.
The tangent lines from P to the circle meet the circumference in two points A and B.
We know that the area of $PAOB$ is equal to $\sqrt{3}\,\text{cm}^2$. What is the distance between P and O?

A. $3\,\text{cm}$

B. $2\,\text{cm}$

C. $4\,\text{cm}$

D. $\frac{\sqrt{3}}{2}\,\text{cm}$

E. $\frac{3}{2}\,\text{cm}$